

Contact Information	Department of Mathematics Cornell University 310 Malott Hall Ithaca, NY 14853-4201 email: vm429@cornell.edu
Professional Appointments	H.C. Wang Assistant Professor (July 2025 – June 2028) Cornell University Mentors: Camil Muscalu, Xiaodong Cao
Education	Ph.D in Mathematics August 2019 – May 2025 University of Maryland, College Park, USA Advisor: Yanir A. Rubinstein Thesis title: From Bergman kernels to Polarity: Perspectives on the Mahler, Bourgain, and Blocki Conjectures MASt in Mathematics October 2017 – June 2018 University of Cambridge, UK Part III of the Mathematical Tripos B.A. Mathematics (4-year degree) October 2013 – July 2017, National and Kapodistrian University of Athens, Greece GPA: 9.74/10
Books	V. Mastrantonis, Y.A. Rubinstein, <i>An Invitation to Convex, Complex, and Differential Geometry via the Mahler conjectures</i> , submitted, solicited by Amer. Math. Soc.
Publications	V. Mastrantonis, <i>L^p-Legendre transforms and Mahler integrals: Asymptotics and the Fokker–Planck heat flow</i> , 2024, arxiv: 2411.10439, to appear in J. Anal. Math. V. Mastrantonis, Y.A. Rubinstein, <i>Two-dimensional Blocki, L^p-Mahler, and Bourgain conjectures</i> , preprint, 2024, arxiv: 2401.10992, to appear in Indiana Univ. Math. J. V. Mastrantonis, <i>A Santaló inequality for the L^p-polar</i> , Contemp. Math. 810 (2025), 31–52. B. Berndtsson, V. Mastrantonis, Y.A. Rubinstein, <i>L^p-polarity, Mahler volumes, and the isotropic constant</i> , Analysis & PDE 17 (2024), 2179–2245. V. Mastrantonis, Y.A. Rubinstein, <i>The Nazarov proof of the non-symmetric Bourgain–Milman inequality</i> , Indiana Univ. Math. J. 73 (2024), 911–953. N. Georgakopoulos, V. Mastrantonis, V. Nestoridis, <i>Relations of the spaces $A^p(\Omega)$ and $C^p(\partial\Omega)$</i> , Results Math. 73 , 86 (2018), 13pp. V. Mastrantonis, C. Panagiotis, <i>Nowhere differentiable functions of analytic type on products of finitely connected planar domains</i> , Monatsh. Math. 187 (2018), 327–341.

Awards

Ivo and Renata Babuska Student Award for Graduate Research for an outstanding Ph.D. student dissertation, April 2026.

AMS–Simons Travel Grant for the academic years 2025–2026 and 2026–2027.

Ann G. Wylie Dissertation Fellowship Award, “Providing full-time support for the latter stages of dissertation writing”, University of Maryland, Fall 2024.

Gerondelis Foundation Graduate Grant, Summer 2024.

Mark E. Lachtman Award, University of Maryland, Spring 2024.

Graduate School Summer Research Fellowship, “Awarded for research support during summer”, University of Maryland, Summer 2023.

Mark E. Lachtman Award, University of Maryland, Spring 2023.

Outstanding Graduate Assistant Award, “Recognized among the top 2% of campus Graduate Assistants for the academic year”, University of Maryland, 2022–2023.

Hauptman Summer Research Award, “Awarded for research support during summer”, University of Maryland, Summer 2022.

George and Marie Vergottis Scholarship, “For studies at the University of Cambridge”, Cambridge Trust, 2017–2018.

Onassis Scholarship, “A competitive scholarship supporting my studies at the University of Cambridge”, Alexander S. Onassis Public Benefit Foundation, 2017–2018.

Pantias–Rallis Endowment Scholarship, “Awarded for excellent academic performance at the Department of Mathematics”, University of Athens, 2014–2015 & 2015–2016.

Teaching experience

Postdoctoral researcher, Cornell University

MATH 7110: Topics in Analysis, Spring 2026 (instructor)

MATH 1110: Calculus I, Fall 2025 (instructor)

Putnam preparation sessions, Fall 2025

For the instructor position, responsibilities include preparing and delivering three hours of lectures per week, assigning and grading homework, holding weekly office hours, and preparing and grading exams.

Teaching assistant, University of Maryland

MATH 463: Complex Variables, Summer 2022 (grader)

MATH 241: Calculus III, Spring 2022 (teaching assistant)

MATH 730: Fundamental concepts of topology, Fall 2021 (grader)

MATH 436: Differential geometry of curves and surfaces, Fall 2021 (grader)

MATH 240: Linear algebra, Spring 2021 (teaching assistant)

MATH 241: Calculus III, Fall 2020 (teaching assistant)

MATH 241: Calculus III, Spring 2020 (teaching assistant)

MATH 141: Calculus II, Fall 2019 (teaching assistant)

Responsibilities included: Leading weekly discussion sessions, preparing and grading homework assignments and in-class worksheets, holding weekly office hours, grading exams, helping students with coursework.

**Outreach and
broader impact**

Co-organized Working Analysis Seminar, Cornell University, Fall 2025–Spring 2026

Mentored five undergraduate students over three years for the **Directed Reading Program** at the University of Maryland:

- Alejandro Escoto on “Geometric Flows in Python”, Fall 2023
- Brooke Guo on “The isoperimetric inequality from antiquity to Steiner”, Spring 2023
- Alejandro Escoto on “The de Rham cohomology”, Fall 2022
- Abdulrahman Alenazi on “The 2-dimensional Mahler conjecture”, Spring 2022
- Aryan Kaul on “The 2-dimensional Brouwer fixed point theorem”, Spring 2021

Gave a talk at the **Interaction between Convex Geometry and Complex Geometry** REU, University of Maryland, July 2024.

Co-organized Research Interactions Team on “Introduction to Complex Geometry” at University of Maryland, Spring 2020

Mentored four students on a paper I coauthored on nowhere differentiable functions within the course “Topics in Analysis”, University of Athens, Spring 2017

Refereed papers for *Geometriae Dedicata*, *Revista Matemática Iberoamericana*, *Advances in Mathematics*, *International Mathematics Research Notices*, and *Mathematische Zeitschrift*.

**Conference and
Workshop Talks**

Third Congress of Greek Mathematicians, Athens, Greece, June–July 2026.

19th Panhellenic Conference on Mathematical Analysis, NTUA, December 2025.

Midwestern Workshop on Asymptotic Analysis, Purdue University Fort Wayne, October 2025.

11-week mini course on the Busemann–Petty problem, Working Analysis Seminar, Cornell University, September–October 2025

Midwestern Workshop on Asymptotic Analysis, Indiana University Bloomington, October 2024.

Göteborg Complex Geometry, Göteborg, Sweden, August 2024.

Recent Developments in Geometric Analysis, AMS Spring Eastern Sectional Meeting at Howard University, April 2024.

8-lecture graduate mini-course, University of Maryland, February–March 2024.

Convex and Complex: Perspectives on Positivity in Geometry, Cetraro, Italy, November, 2022.

8-lecture graduate mini-course, University of Maryland, October–November 2021.

Seminar Talks

Geometric and Functional inequalities and Applications, May 2025

Analysis Seminar, Cornell University, May 2025

Analysis Seminar SAMPS, NTUA, January 2025

Measure Theory Seminar, Kent State University, March 2024.